

Slot 12 · Write tests that fail

File: `test_model.py` · **Done when:** `uv run pytest tests/test_slot12.py` shows 4 passed · **Time:** 18 min

Today's idea

For two days, green meant good. **Today you score a point for every test the model FAILS.** These are not accuracy tests. They are **behaviour** tests — things any sensible ETA model must do, whatever its MAE.

kind	rule	example
invariance	change something that should not matter → answer must not move	a different <code>order_id</code>
directional	change something that matters → answer must move the right way	double the distance → not faster
slice	no group of orders may be much worse than overall	long trips, night, Pune

Step 1 · Look at the ready-made parts

Open `test_model.py`. Already there for you:

- `bundle` — the model the service uses
- `test_rows` — 4,000 orders the model has never seen
- `predict(bundle, rows)` — returns an array of ETAs
- one finished example: `test_order_id_does_not_matter`

Step 2 · Write at least four tests

Start with the two unfinished ones, then add your own. The pattern for a directional test:

```
def test_more_distance_never_faster(bundle, test_rows):
    a = predict(bundle, test_rows)
    b = predict(bundle, test_rows.with_columns(
        (pl.col("distance_km") * 2).alias("distance_km")))
    bad = int((b < a - 0.01).sum())          # 0.01 = small tolerance
    assert bad == 0, f"{bad}/{len(a)} rows got FASTER when distance doubled"
```

More ideas:

- heavy rain never faster than clear (set `weather` to `"clear"`, then to `"heavy_rain"`)
- 4 more items never faster
- every prediction between 3 and 240 minutes
- MAE on long trips (`distance_km > 10`) at most $1.25 \times$ the overall MAE

```
uv run pytest test_model.py -v
```

Red is good today. Count your failures.

Step 3 · Reference results

On the shipped model, a good set of tests gives **4 failed, 5 passed**:

test	result
doubling distance never faster	FAIL — 15 of 4,000 rows got faster
heavy rain never faster than clear	FAIL — 55 of 4,000 rows got faster
4 more items never faster	FAIL — 6 of 4,000 rows got faster
long trips > 10 km	FAIL — MAE 7.58 vs 5.60 overall (1.35×)

Think For 55 orders, making the weather **worse** gives a **faster** ETA. Is that noise? No metric caught it. Would a customer?

Step 4 · The reveal (together, after everyone has written tests)

```
uv run python scripts/slice_table.py
```

Look at the **bias** column, not the MAE column. Every slice is about **−3.5 minutes**: the model is too optimistic by the same amount everywhere. Then read the last three lines — what happened around **day 274**? (Remember the 76.5% from Slot 4?)

Step 5 · Check you are done

```
uv run pytest tests/test_slot12.py
```

This checks that you wrote at least 4 tests, no `NotImplementedError` is left, and that **some of your tests fail**. If all of them pass, your tests are too easy!

If something goes wrong

what you see	what to do
every test passes	Your tests are too gentle. Try bigger changes, or slices.
every test fails	You compared floats exactly. Use a tolerance: <code>b < a - 0.01</code> .
error about changing a DataFrame	polars tables do not change in place. Use <code>with_columns</code> .
<code>test_nothing_is_still_a_stub</code> fails	Remove every <code>raise NotImplementedError</code> .